

Date	03 July 2026
Team ID	SWTID-2026-4593
Team Size	3
Team Leader	Hari Bellamkonda
Team Member 1	Kesha Vardhan Chinni
Team Member 2	Akili Venkata Siva Sai Srikar
Project Name	Machine Learning-Based Credit Card Approval Prediction System
Maximum Marks	4 Marks

Sample Project Documentation

Introduction

The Sample Project Documentation provides a comprehensive overview of the Machine Learning-Based Credit Card Approval Prediction System. It summarizes the project objectives, development methodology, software architecture, implementation approach, technologies used, and expected outcomes. This document serves as a reference for understanding the overall design and execution of the proposed intelligent credit card approval prediction solution.

Project Overview

Project Title	Machine Learning-Based Credit Card Approval Prediction System
Project Domain	Artificial Intelligence and Machine Learning
Application Type	Web-based Intelligent Prediction System
Programming Language	Python
Framework	Flask
Database	MySQL
Machine Learning Libraries	Scikit-learn, Pandas, NumPy
Development Tools	Visual Studio Code, Jupyter Notebook
Deployment Platform	Render with Docker

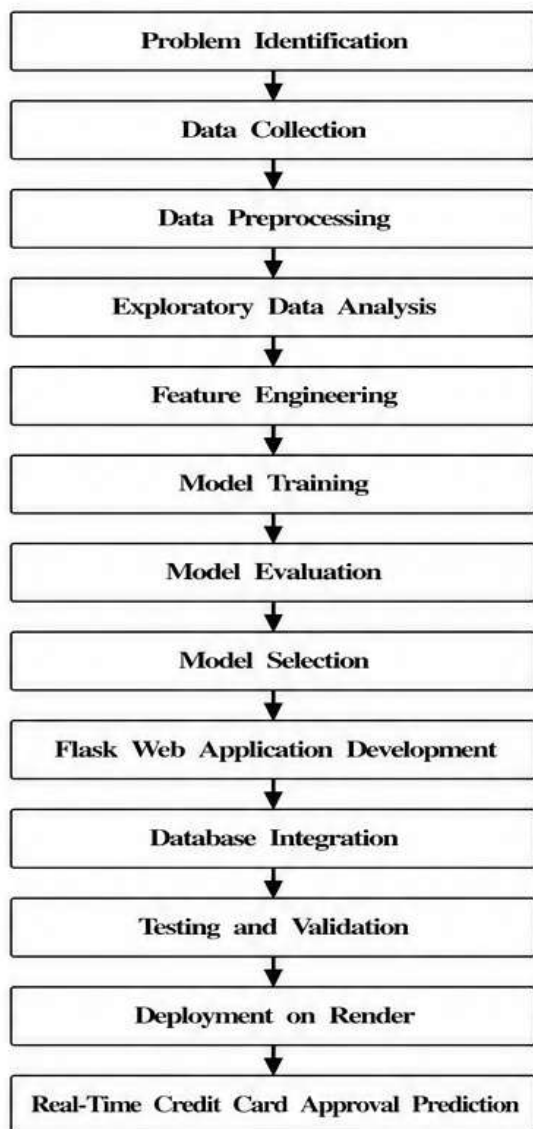
Project Objectives

- Automate the credit card approval process.
- Reduce manual verification effort.
- Improve decision-making accuracy using Machine Learning.
- Provide real-time approval prediction.
- Develop an intuitive web application.
- Maintain secure applicant information.
- Support scalable deployment.
- Improve operational efficiency for financial institutions.

Major Modules

Module	Description
User Authentication	Secure registration and login.
Applicant Management	Collects applicant information.
Data Preprocessing	Prepares data for prediction.
Machine Learning Prediction	Predicts approval or rejection.
Database Management	Stores applicant records.
Result Dashboard	Displays prediction results.

Project Workflow



Project Features

- Secure user authentication.
- Applicant profile management.
- Intelligent credit approval prediction.
- Multiple Machine Learning algorithms.
- Responsive web interface.
- Database integration.
- Prediction history management.
- Real-time decision support.

Expected Outcomes

Outcome	Description
Accurate Prediction	Improves approval decision accuracy.
Automation	Reduces manual processing effort.
Efficiency	Provides faster application processing.
Security	Protects applicant information.
Scalability	Supports future system expansion.
Maintainability	Simplifies future enhancements.

Technology Summary

Frontend	HTML5, CSS3, JavaScript, Bootstrap 5
Backend	Python, Flask
Machine Learning	Scikit-learn, Pandas, NumPy
Database	MySQL
Deployment	Render and Docker
Version Control	Git and GitHub

Conclusion

The Machine Learning-Based Credit Card Approval Prediction System demonstrates how Artificial Intelligence can automate financial decision-making through accurate prediction, secure data management, and an intuitive web application. The project integrates modern Machine Learning techniques with web technologies to deliver a scalable, reliable, and user-friendly solution for credit card approval prediction.